

AI-compute supply chain: chart pack

Current state of the analysis. Data: UN Comtrade + TDM monthly panel (60 HS6 codes, 2017-01..2026-04, audited vs Atlas) and Atlas HS2012 annual. Sections: 1 the chain as flows (2024) -- 2 the time-varying factor model -- 3 concentration & fragmentation. Details: docs/data.md, docs/supply-chain-narrative.md, results/.

2026-07-27

1. The chain as flows

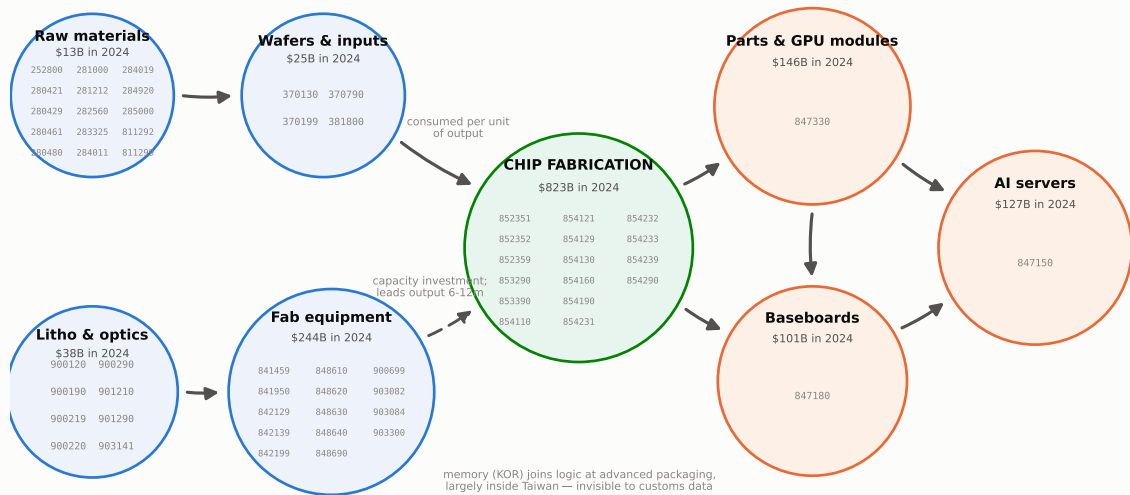
2024 bilateral flows, China+Hong Kong merged (CHK), intra-bloc excluded. Stages 1-5 from Atlas HS2012 annual data; stages 6-8 from the monthly panel (docs/data.md §3).
Generated by scripts/export_supply_chain_sankey.py.

1. The chain as flows

The chain's stylized shape, with HS6 codes per node

Two parallel input branches (materials; tools) converge on chip fabrication; the output side is sequential. Node heights scale with the log of 2024 world trade. Solid arrows: consumed per unit of output; dashed: capacity investment (leads output 6-12 months in fab countries).

The AI-compute supply chain — stylized shape, with the HS6 codes in each node



Two parallel input branches converge on fabrication; the output side is sequential. Circle areas follow the log of 2024 world trade (illustrative, so chips does not dwarf raw materials). Solid arrows: inputs consumed per unit of output. Dashed: capacity investment. Design/EDA/IP value enters as services, never as goods trade. Codes: OECD semiconductor mapping + Fed AI-compute basket (docs/data.md §1).

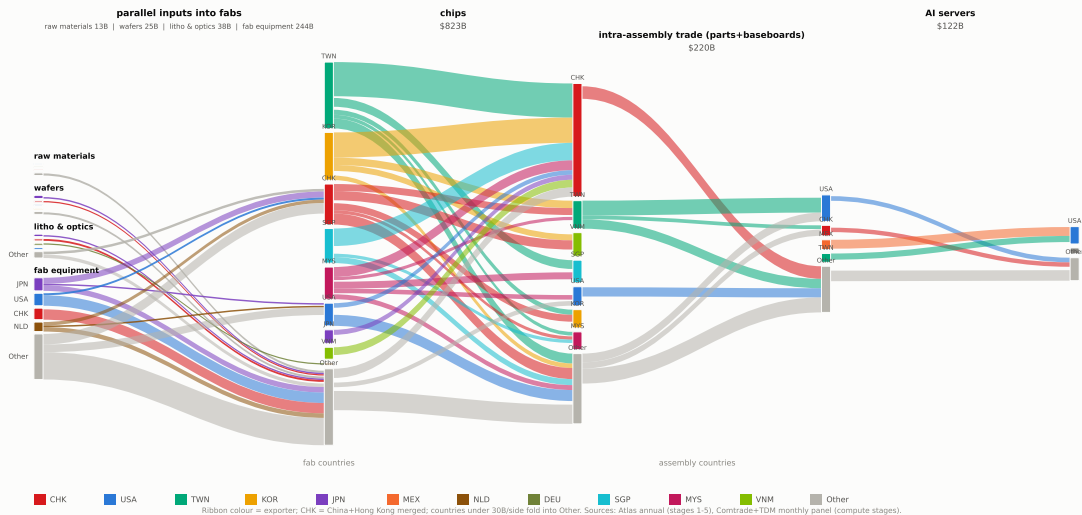
1. The chain as flows

Whole-chain overview, dollar scale (coarse)

All eight stages left to right; band width = 2024 dollars.

Inputs, litho optics and equipment run in parallel and converge on the fab countries; parts and baseboards form the intra-assembly segment before servers.

The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (one dollar scale — band widths nominally comparable everywhere)

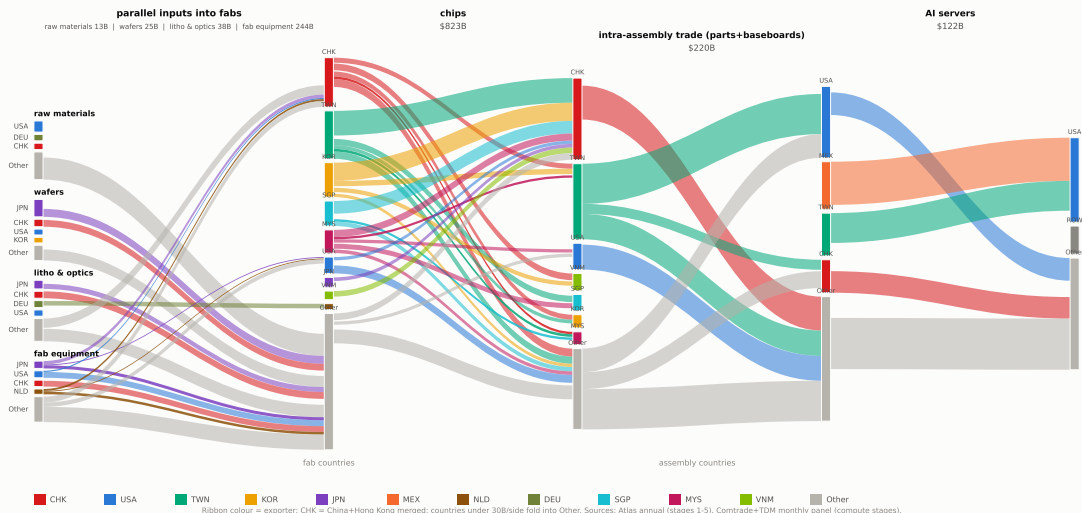


1. The chain as flows

Whole-chain overview, normalized (coarse)

Same flows with each stage rescaled to equal width, so small stages' composition is readable (dollar scale makes chips dwarf everything).

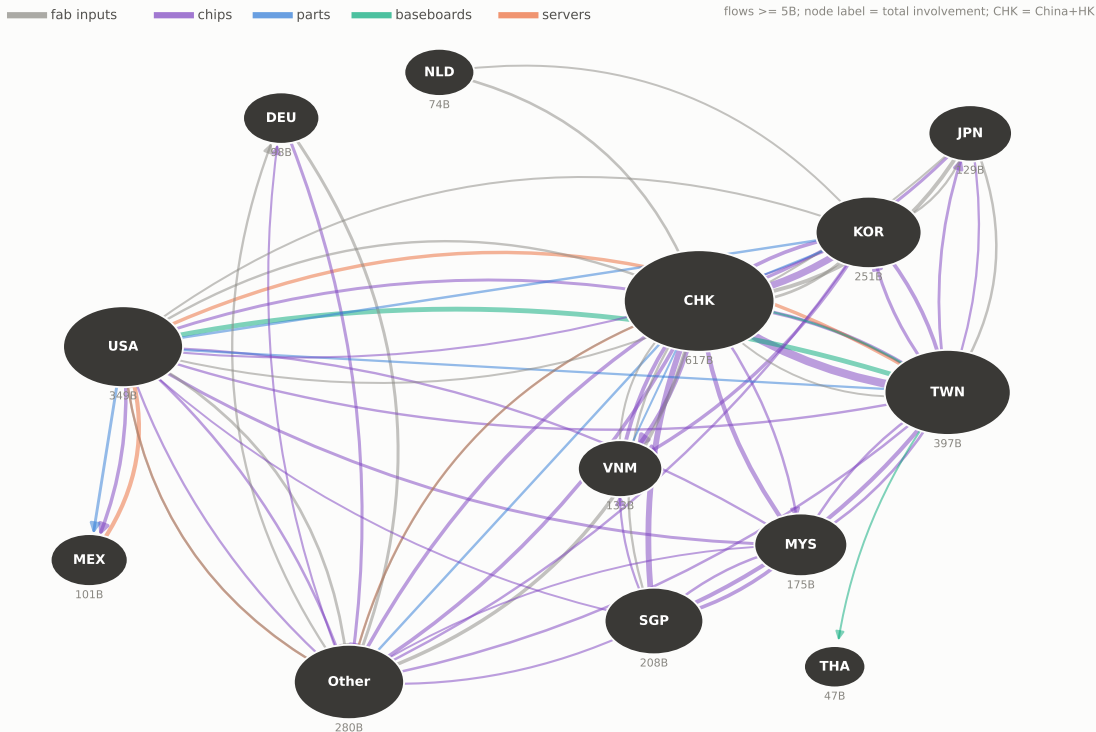
The AI-compute supply chain, 2024 — parallel inputs converge on the fabs; chips flow on to servers (stages equalized — shares only)



Country network, all stages

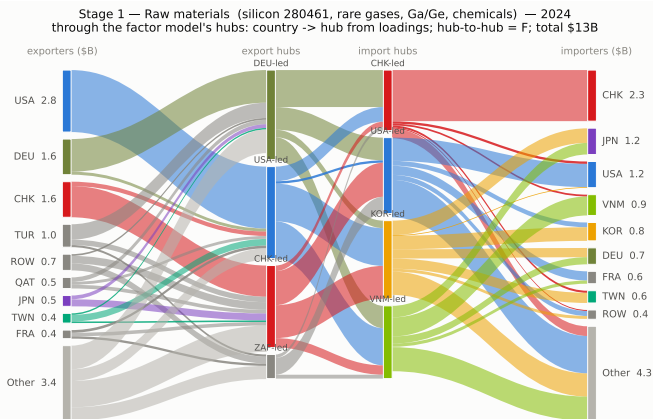
Nodes = countries, edges = 2024 flows colored by stage. Shows integrated roles a stage sequence cannot: Mexico takes in chips and parts and ships out servers at a single node.

The AI-compute supply chain as a country network, 2024 — edges by product stage, width ~ \$



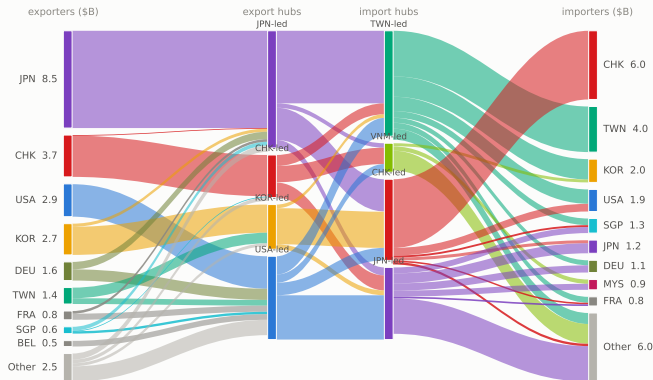
Stage 1 -- raw materials, hub decomposition

Four-column view: exporters -> export hubs -> import hubs -> importers, hubs estimated by MFM in the nonnegativity-identified basis. Applies to every stage page that follows; fits R^2 0.95-0.99.



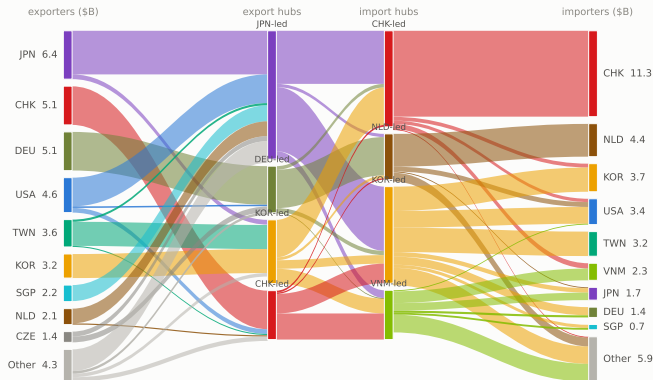
Stage 2 -- wafers & wafer inputs

Stage 2 — Wafers & wafer inputs (381800, 3701xx/370790) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$25B



Stage 3 -- lithography & optics inputs

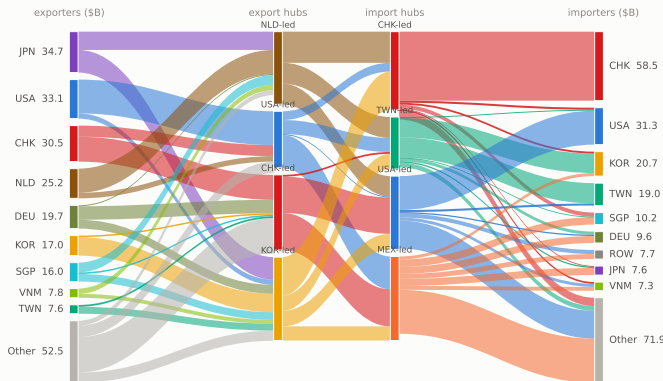
Stage 3 — Lithography & optics inputs (9001xx/9002xx, 901210/90, 903141) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$38B



Stage 4 -- fab equipment

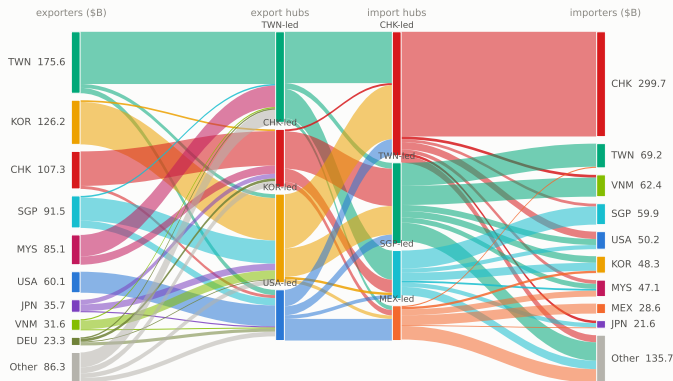
Capacity goods, not consumables: in fab countries these imports lead chip exports by 6-12 months (docs/notes/edge-type-tests.md).

Stage 4 — Fab equipment (8486xx, metrology, fab plant) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$244B

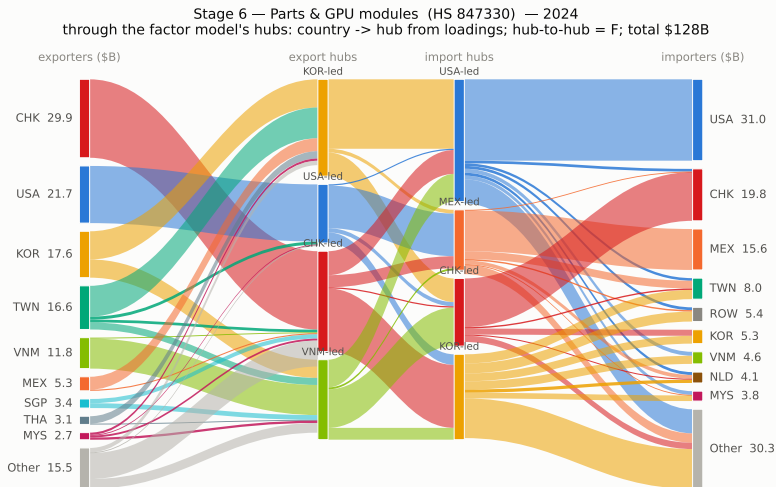


Stage 5 -- chips (logic, memory, discretetes)

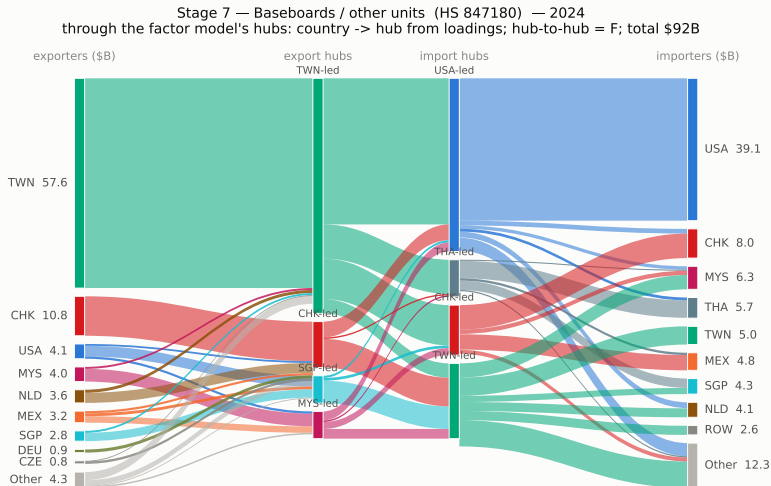
Stage 5 — Chips (logic, memory, discretetes) (8542xx, 8541xx, media) — 2024
through the factor model's hubs: country -> hub from loadings; hub-to-hub = F; total \$823B



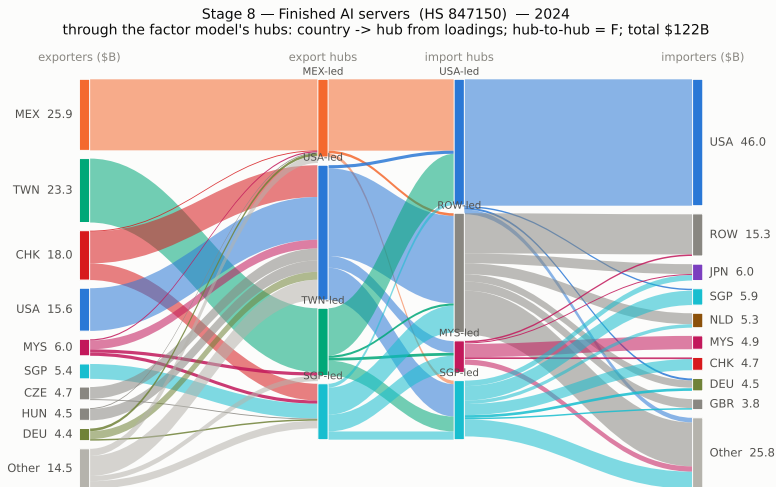
Stage 6 -- parts & GPU modules (847330)



Stage 7 -- baseboards (847180)



Stage 8 -- AI servers (847150)



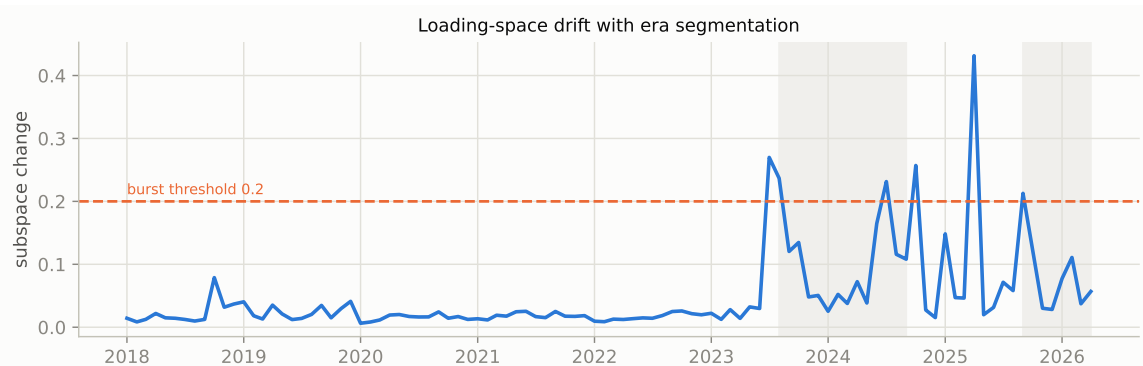
2. The time-varying factor model

Era-anchored TV-MFM on the audited monthly panel (12m trailing window, $k=4$, \$B levels, NNF basis), 2017-12..2026-04. Shaded bands = eras between structural breaks. Generated by `scripts/tvmfm_monthly_anchored.py`; summaries in `results/mfm/tvmfm/`.

2. The time-varying factor model

AI compute, CHK basis -- loading drift and eras

The preferred basis (entrepot churn netted; Taiwan separate).
One unbroken 68-month era from 2017-12, then breaks at
2023-08, 2024-10, 2025-09.

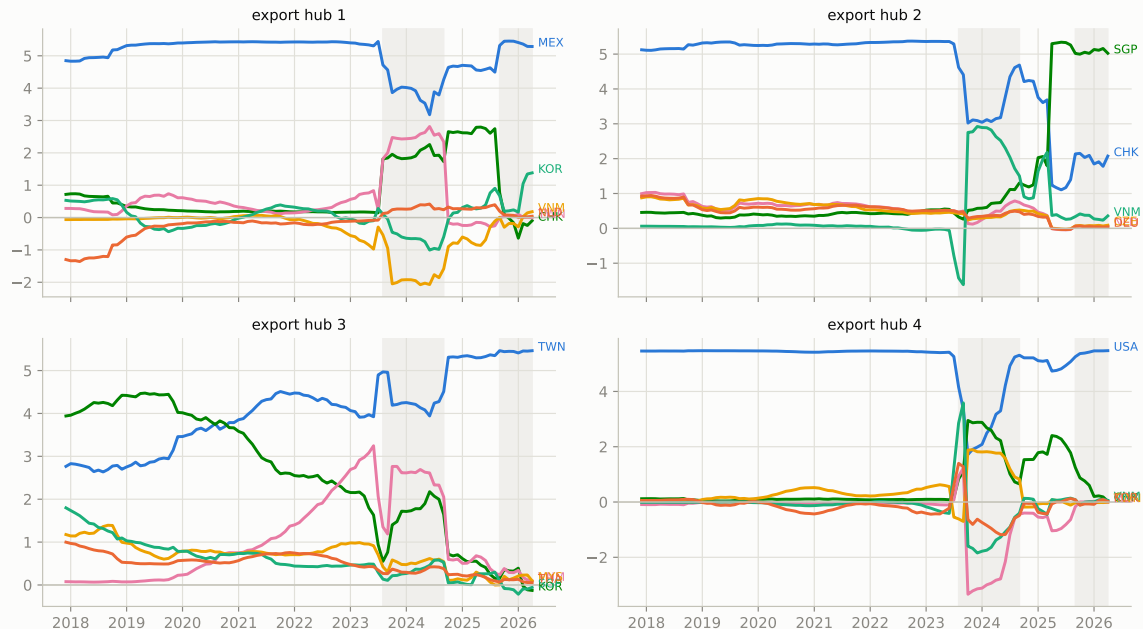


2. The time-varying factor model

AI compute, CHK basis -- export-hub loadings

Per-hub country loadings through time. At 2024-10 the China-bloc hub loses its separate identity (cross-era similarity 0.45) and a Singapore-led hub appears.

AI compute, China+HKG bloc (TWN separate) — export loadings, era-anchored (12m window; shaded bands = eras)

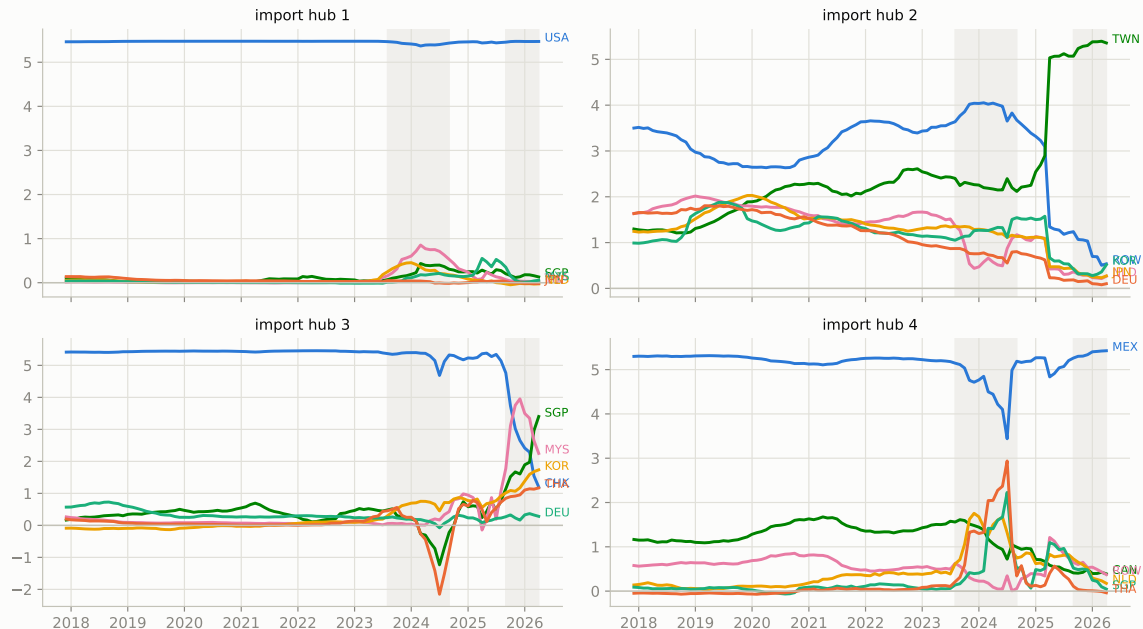


2. The time-varying factor model

AI compute, CHK basis -- import-hub loadings

Import hubs are the stable side of the network (USA, Mexico, Taiwan, rest-of-world led).

AI compute, China+HKG bloc (TWN separate) — import loadings, era-anchored (12m window; shaded bands = eras)

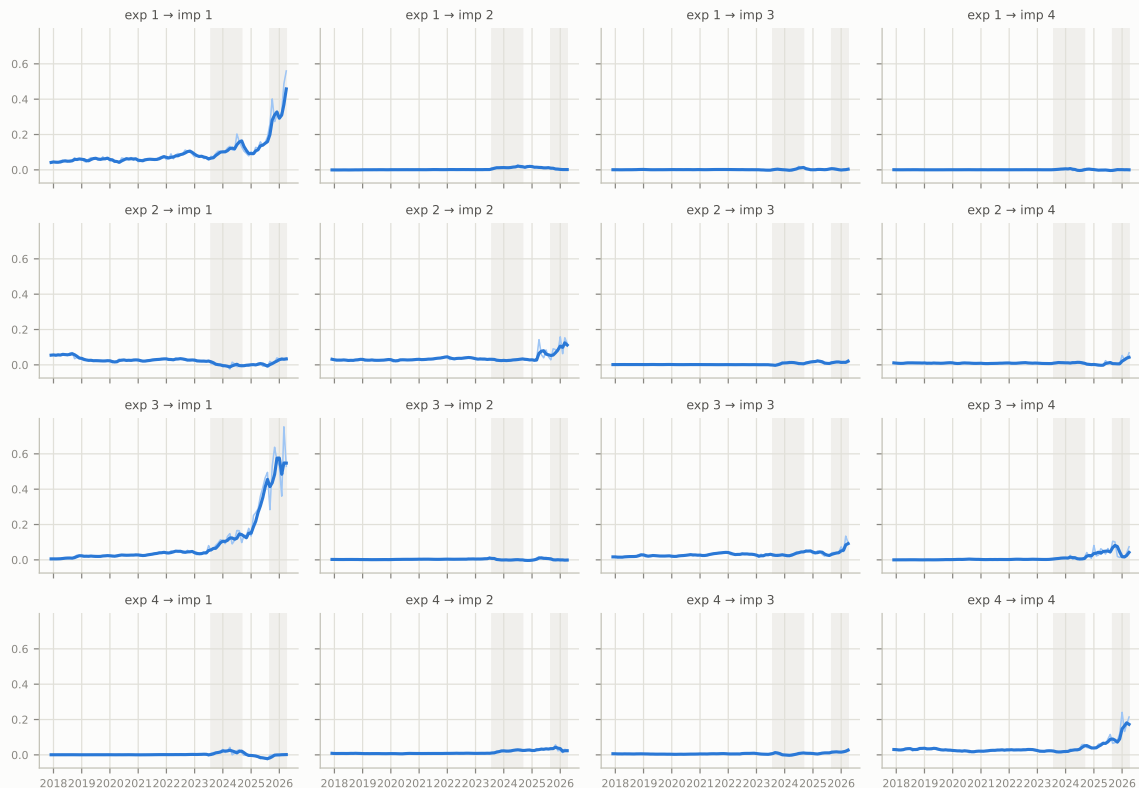


2. The time-varying factor model

AI compute, CHK basis -- hub-to-hub intensities F_t

Each panel is one export-hub x import-hub channel in \$B (3m MA thick).

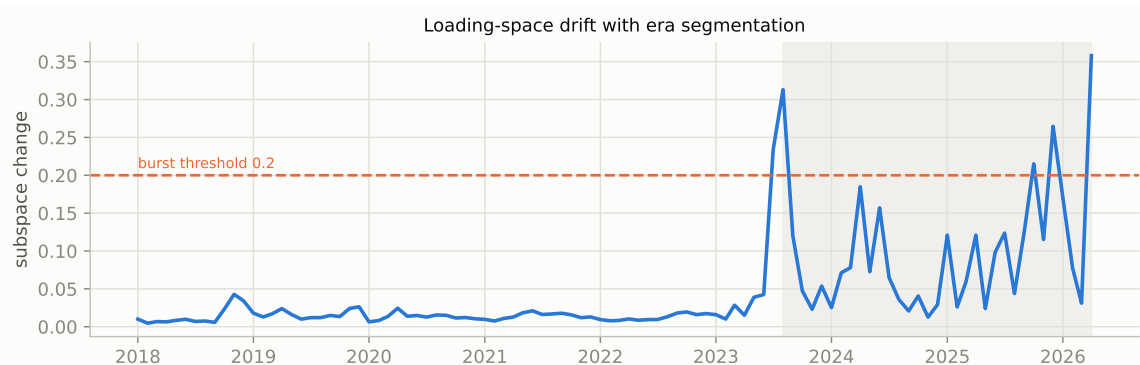
AI compute, China+HKG bloc (TWN separate) — hub-to-hub F_t , era-anchored (3m MA thick)



2. The time-varying factor model

AI compute, country level -- drift

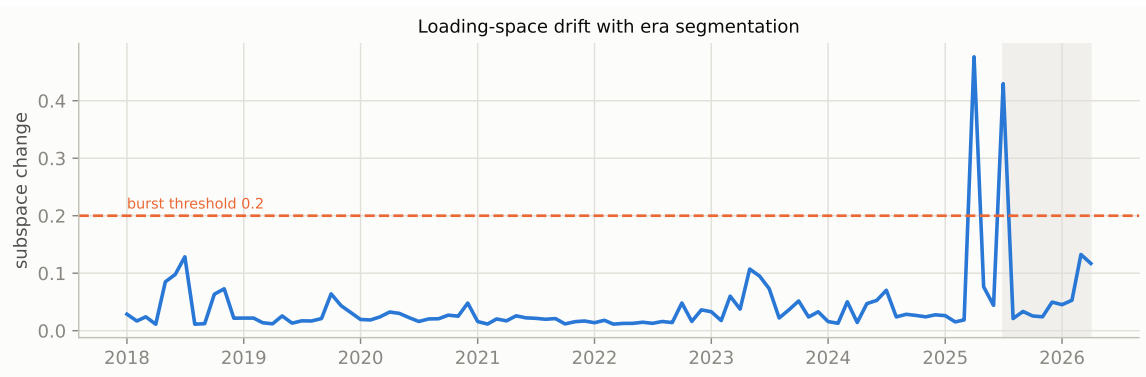
Without bloc merging the sample splits once: 2023-08. The demand event is the only dominant country-level break in nine years.



2. The time-varying factor model

Parts (847330), country level -- drift

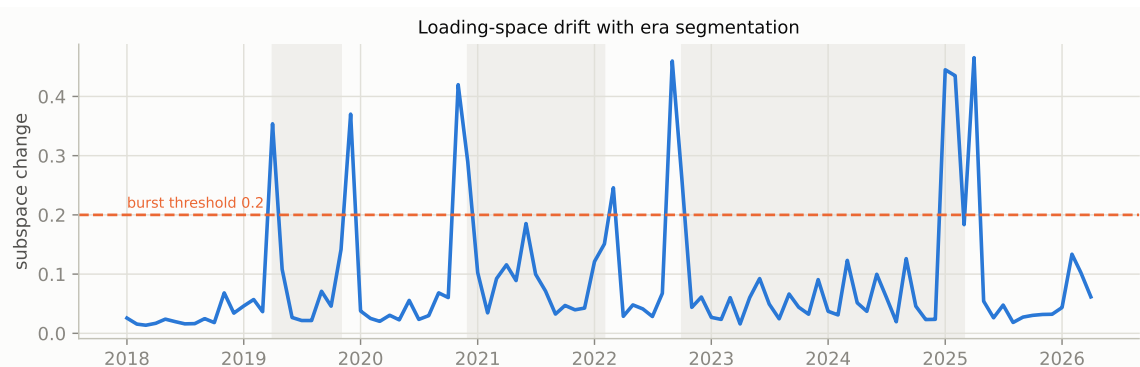
91 months without a structural break; the parts layer is the stable substrate of the network.



2. The time-varying factor model

AI compute, double-bloc basis -- drift

CHK and USA+MEX both merged, isolating cross-bloc structure.
Breaks at 2019-04/2019-12 (first tariff war), 2022-10 (first export controls), 2025-04 (tariffs): policy breaks blocs.

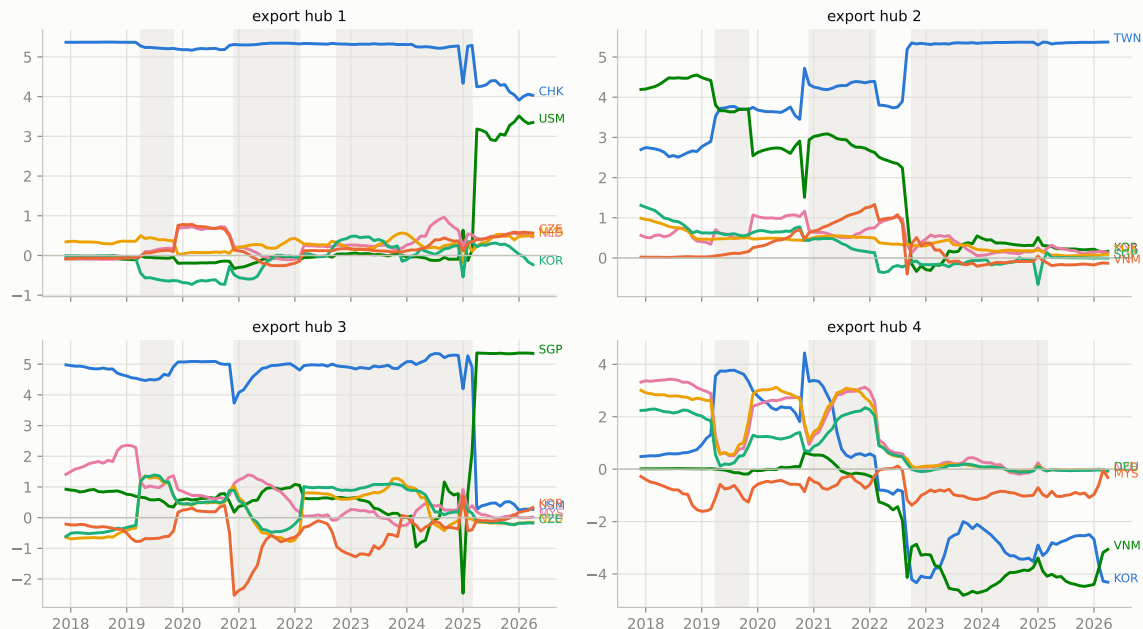


2. The time-varying factor model

AI compute, double-bloc basis -- export-hub loadings

In the 2025-04 era the two blocs load together on one factor (CHK +3.97, USM +3.41) -- the entanglement signature unique to the tariff break.

AI compute, CHN+HKG and USA+MEX blocs — export loadings, era-anchored (12m window; shaded bands = eras)



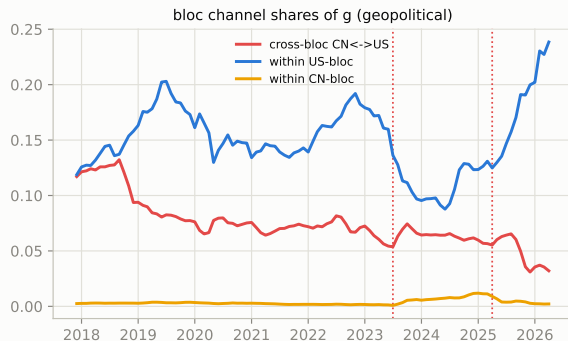
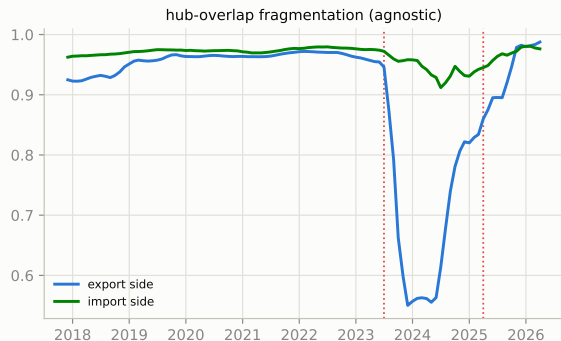
3. Concentration & fragmentation

Measure system of docs/concentration-fragmentation-nnf-mfm-trade.md computed monthly from the TV-MFM (CHK basis).
Dotted lines mark 2023-07 and 2025-04. Interpretation:
results/network_stats/findings.md. Generated by
scripts/network_stats.py.

Two notions of fragmentation -- AI compute

Left: hub-overlap fragmentation (range-compressed under the anchored basis; mainly a diagnostic). Right: bloc channel shares -- the cross-bloc CN-US share falls ~70% over the sample in two legs (2019, then 2023 onward) with a plateau between.

AI compute (3 codes) — two notions of fragmentation (3m MA)



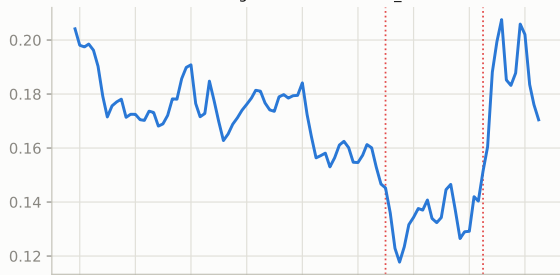
3. Concentration & fragmentation

Joint concentration decomposition -- AI compute

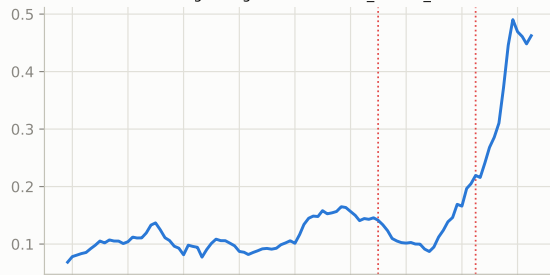
$\text{HHI_flat} = \text{HHI_F} \times (\mu_a \mu_b + \text{alignment covariance})$, exact identity. Alignment covariance is mildly positive from 2017, peaks ~2019, decays through the AI era: no compounding-concentration signature.

AI compute (3 codes) — joint concentration decomposition (3m MA; dotted = 2023-07, 2025-04)

linkage concentration HHI_F



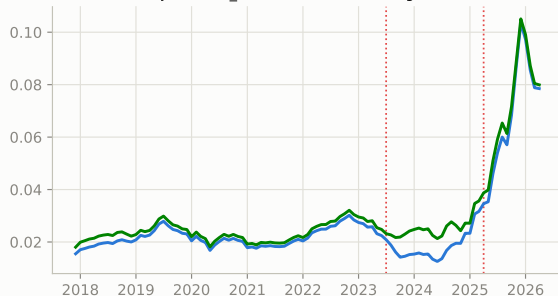
linkage-weighted means $\mu_a * \mu_b$



alignment covariance $\text{Cov}(a,b)$



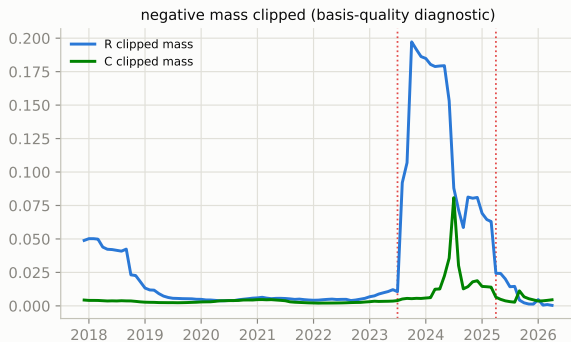
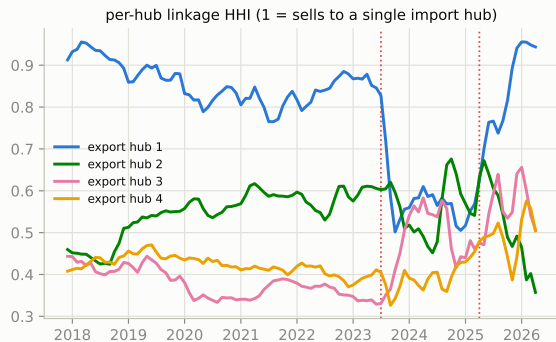
joint: HHI_flat (blue) vs bilateral (green)



Per-hub linkage concentration and basis diagnostic -- AI compute

Left: does an export hub sell into one import program or several. Right: clipped negative mass (basis quality), elevated during the 2023-24 regime transition.

AI compute (3 codes) — linkage structure and basis diagnostics

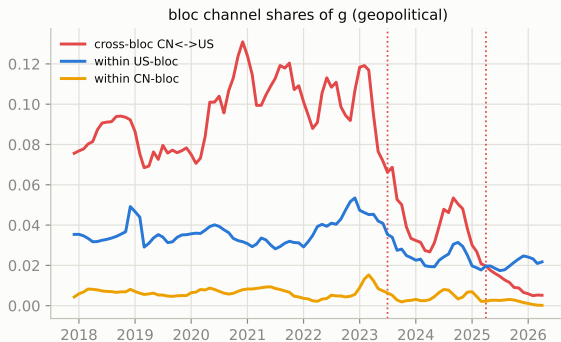
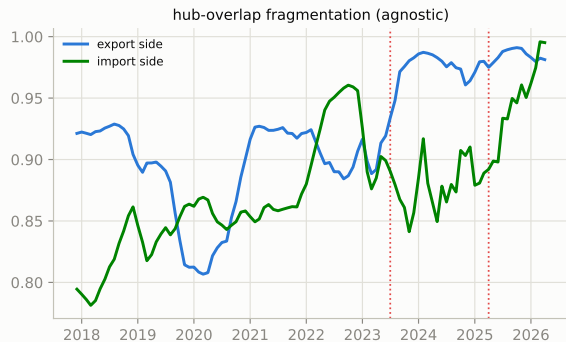


3. Concentration & fragmentation

Fragmentation -- baseboards (847180)

The sharpest decoupling of the three codes: the cross-bloc share collapses ~20x from its 2021 peak while linkage concentration HHI_F rises 0.09 -> 0.48.

HS 847180 (baseboards) — two notions of fragmentation (3m MA)



3. Concentration & fragmentation

Fragmentation -- parts (847330)

Cross-bloc share halves but remains the highest of the three codes: the parts layer stays the shared substrate.

HS 847330 (parts/GPU cards) — two notions of fragmentation (3m MA)

